

Modular 5-DOF Articulated Robotic Arm with Automatic End-Effector Changing System

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The University of Texas
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*Manufacturing &
Industrial Engineering*



In Collaboration With:



Project Objectives

Management

- Design and development of a 5-DOF articulated robotic arm
- Automatic end-effector swapping mechanism
- Automated pick-and-place operations
- Constrained to fit within custom glovebox dimensions

Product Specifications

1.00 KG

Payload

50.00 CM

Total Reach

600 WATTS

Power Consumption

24 VOLTS

Required Voltage

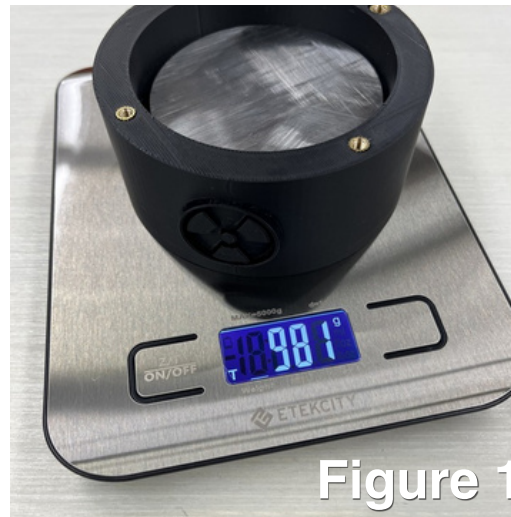


Figure 1

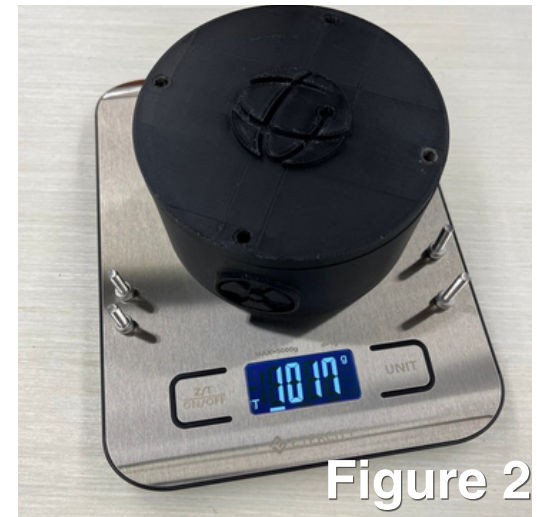
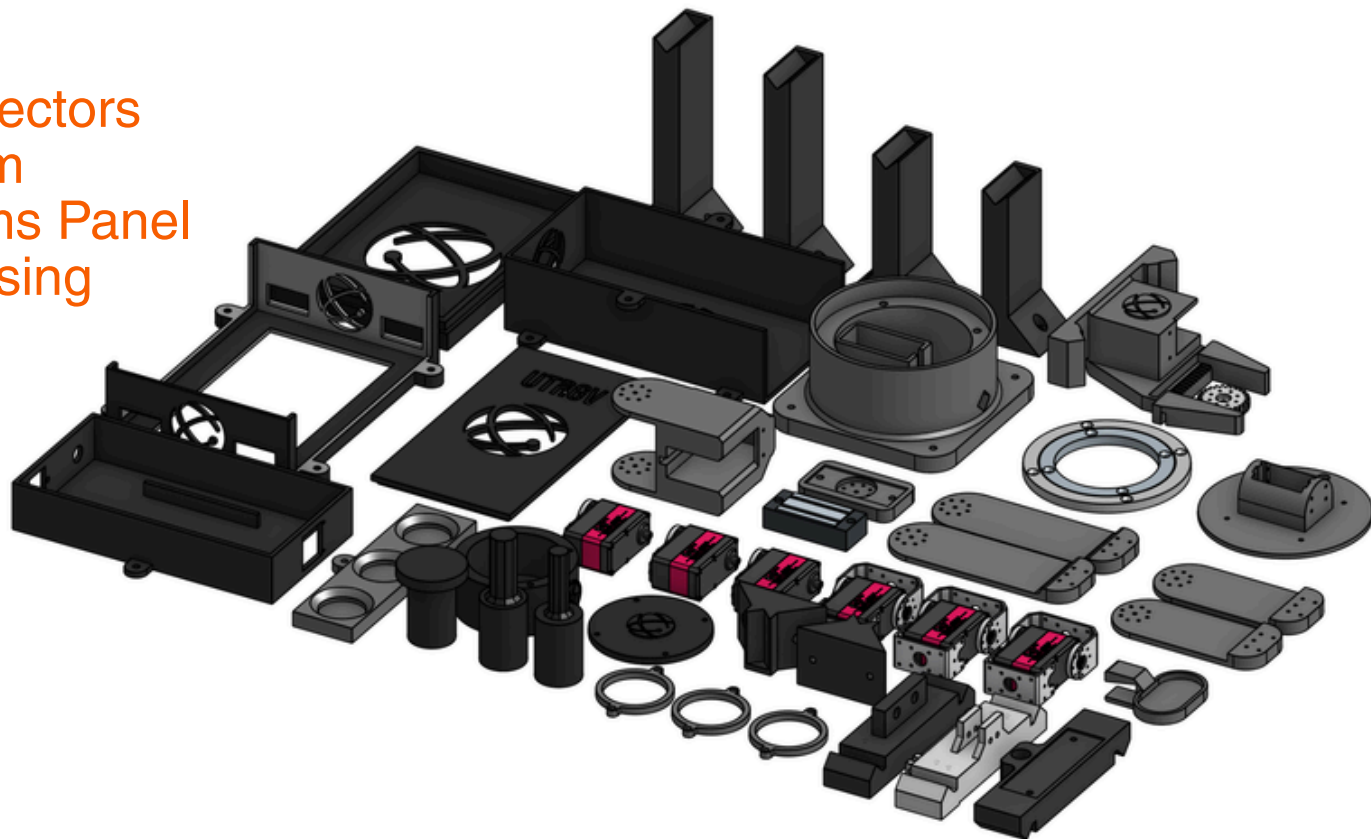


Figure 2

Solution Development - Subsystems

Systems

- Base Assembly
- Linkage System
- Custom End-Effectors
- Coupling System
- Auxiliary Systems Panel
- Electronics Housing
- GUI interface

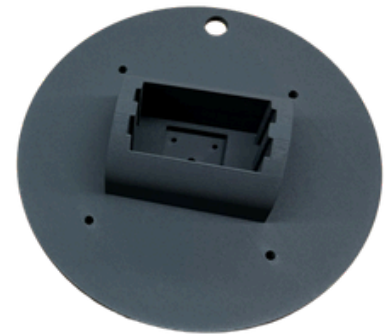
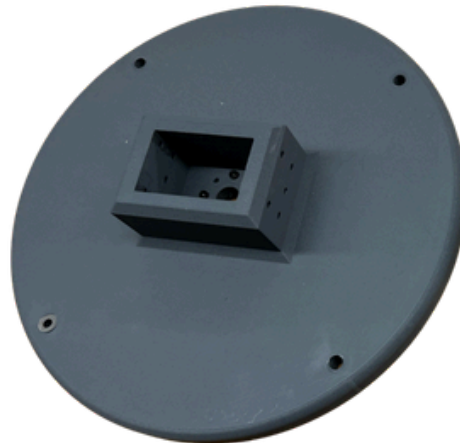
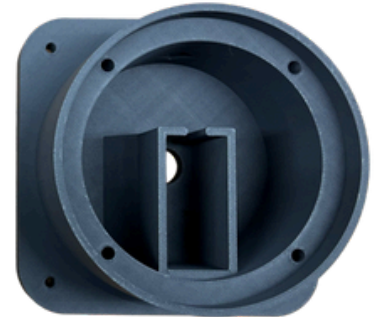
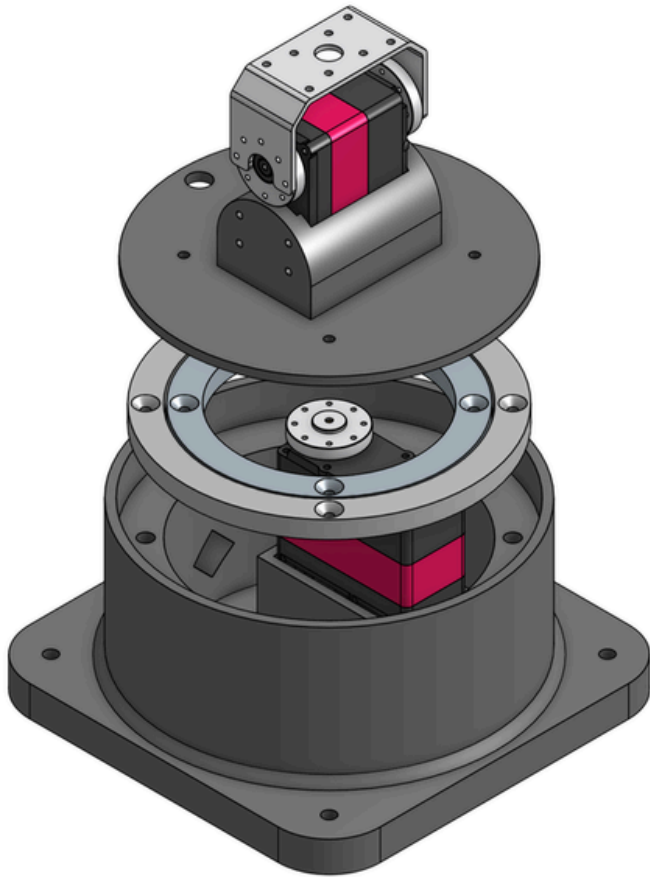


In Collaboration With:



Base Design

Prototype Iteration



Before

After

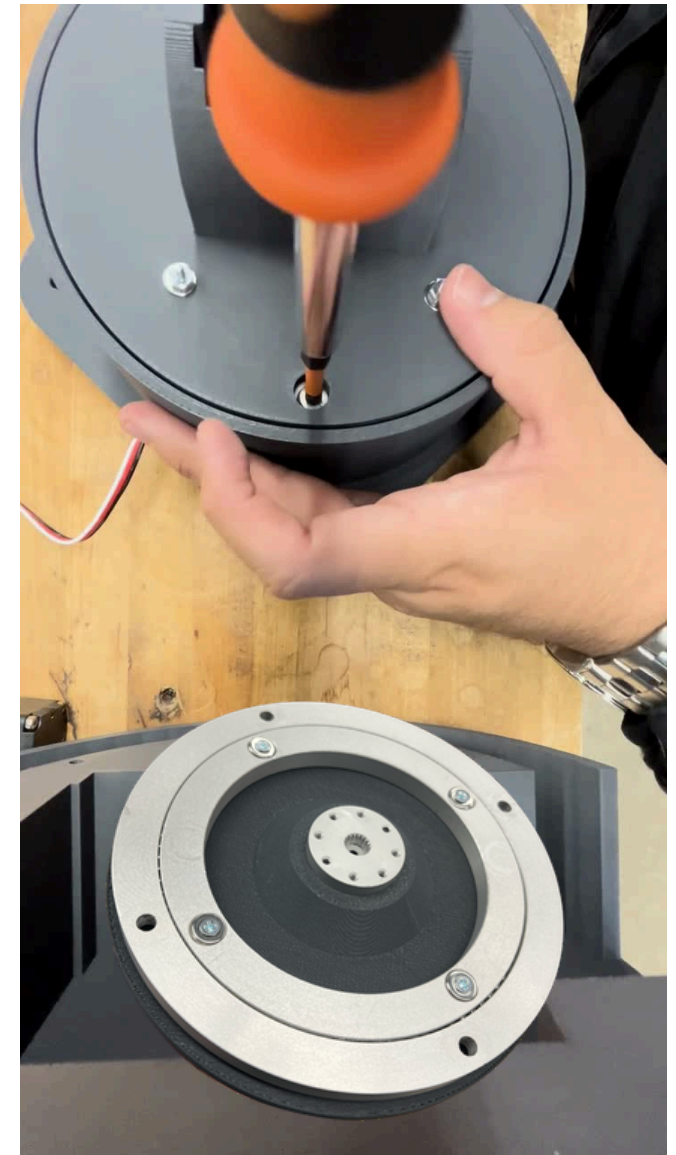
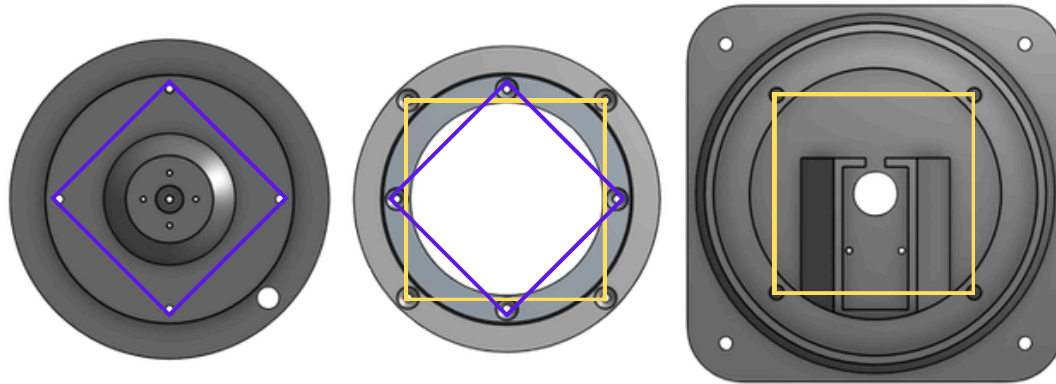
In Collaboration With:



Base Assembly

DFA

- Fastener Access
- Assembly Sequence
- Reduced Handling
- Service Access
- Part Alignment
- Maintainability

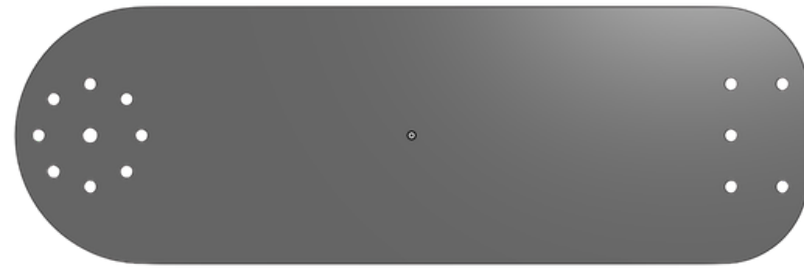


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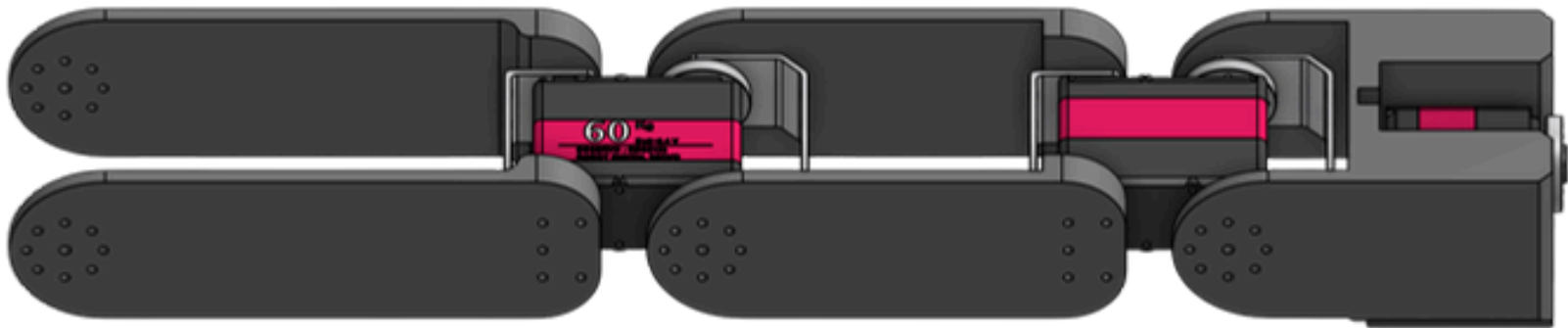


Linkage System

Reach vs. Torque



Link	Length (mm)
Link 1	185
Link 2	150
Link 3	115



In Collaboration With:



Custom End-Effectors

Modularity

- Servo Gripper
- Pneumatic Gripper
- Electric Pump

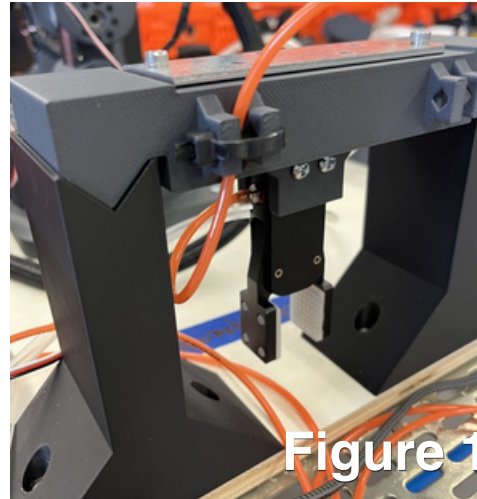
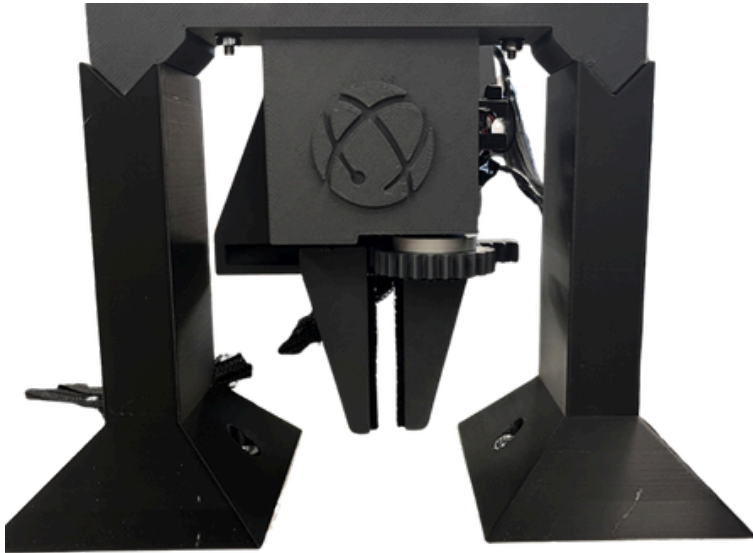


Figure 1

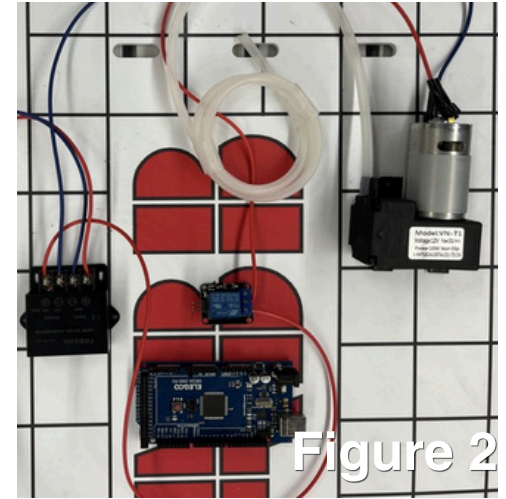


Figure 2

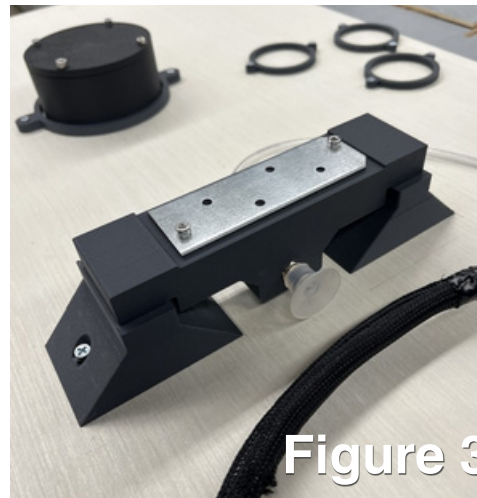


Figure 3

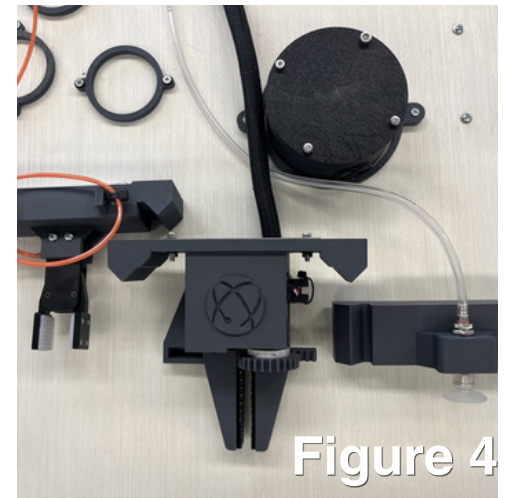


Figure 4

In Collaboration With:



Coupling System

Electromechanics

- Electromagnet
- 120LB (60KG) Rating
- Relay-Controlled

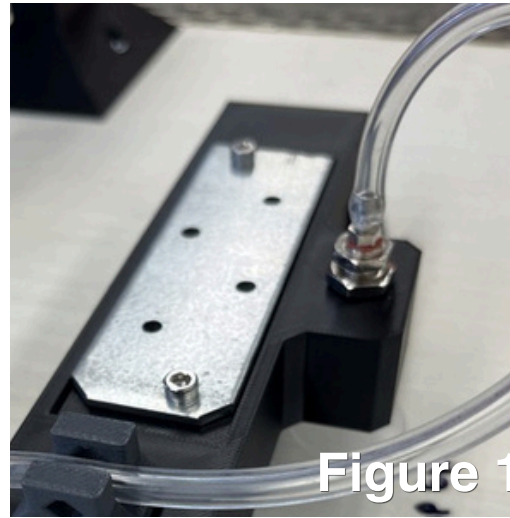


Figure 1

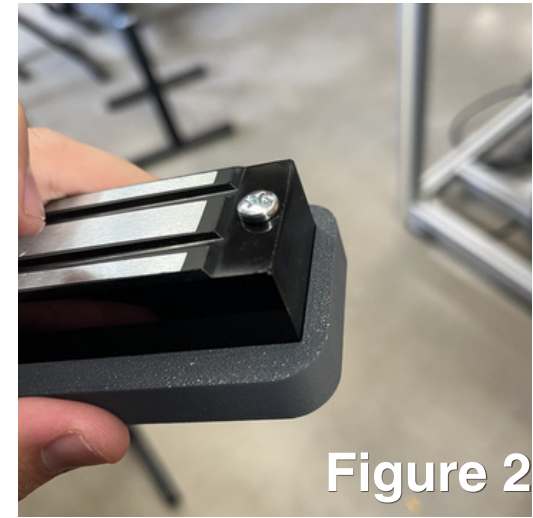
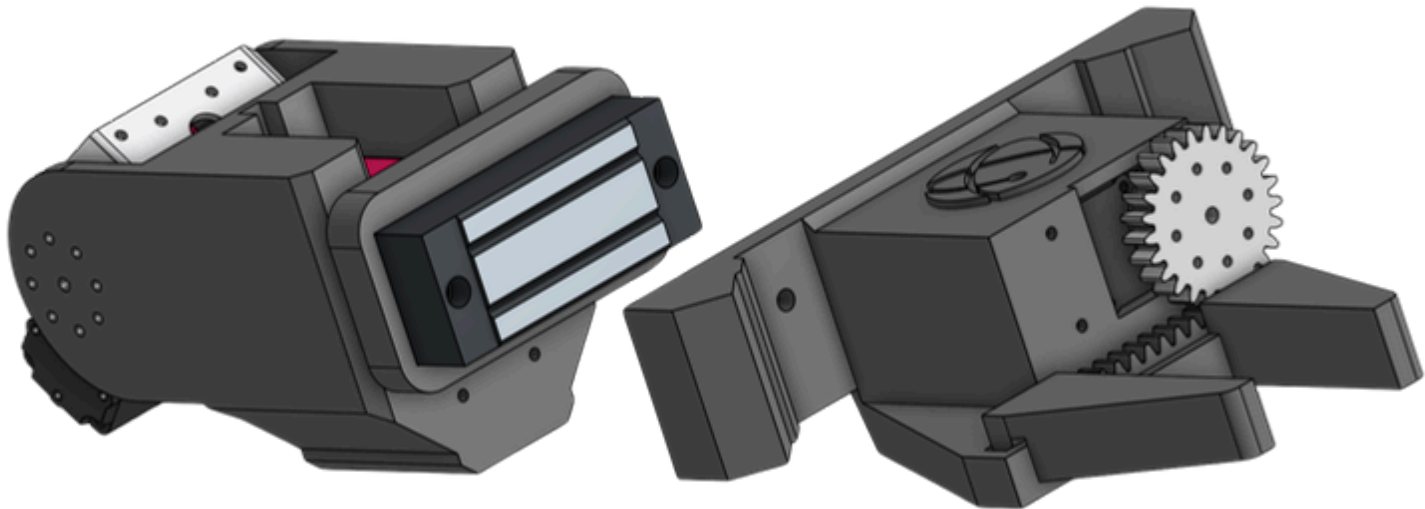


Figure 2



In Collaboration With:



DFM for 3D Printing

Manufacturing Constraints

- Support Strategy
- Support Strength
- Print Orientation
- Assembly Fit
- Build Limits
- Material Limits

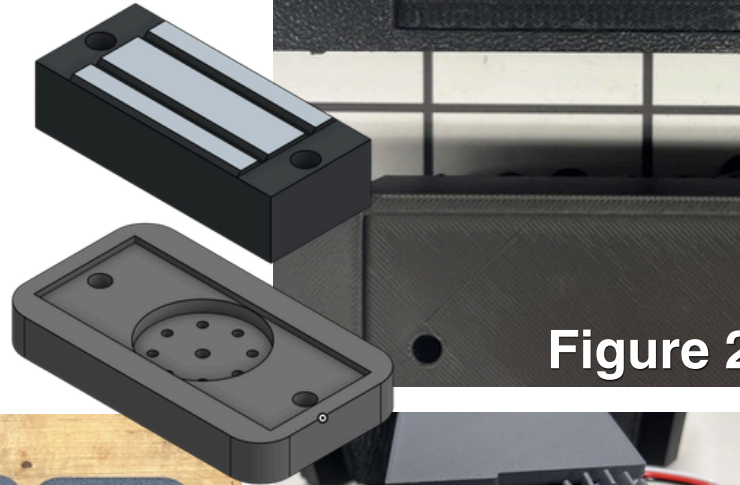


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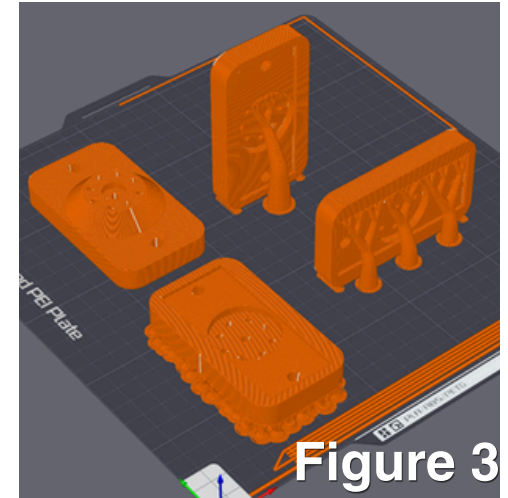


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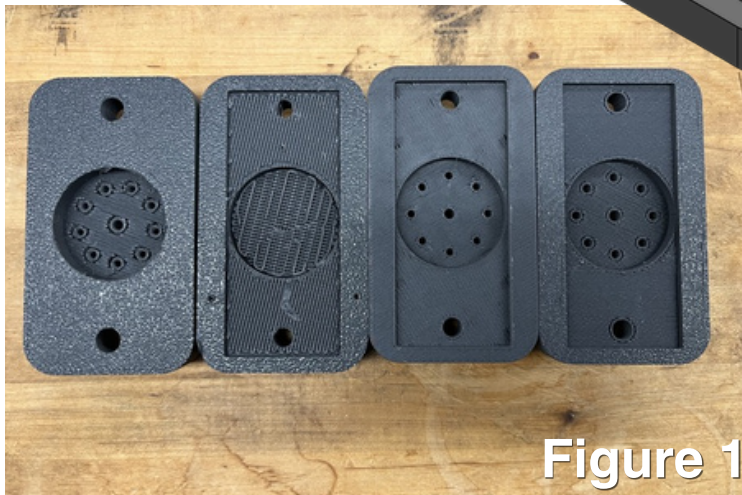


Figure 1

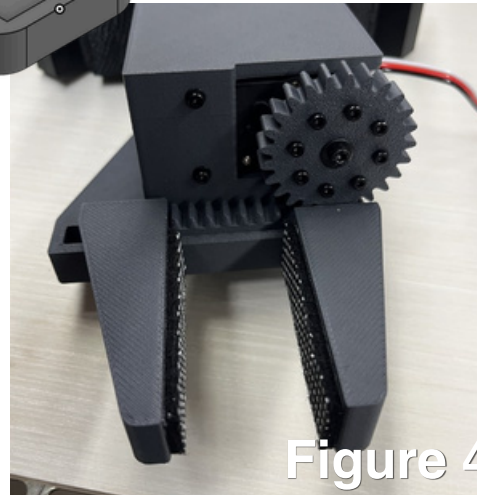


Figure 4

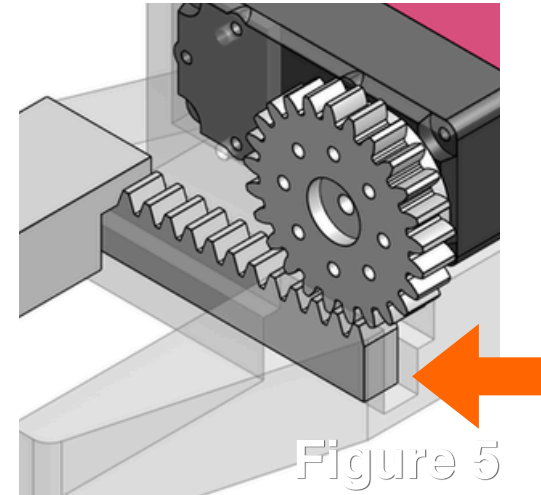


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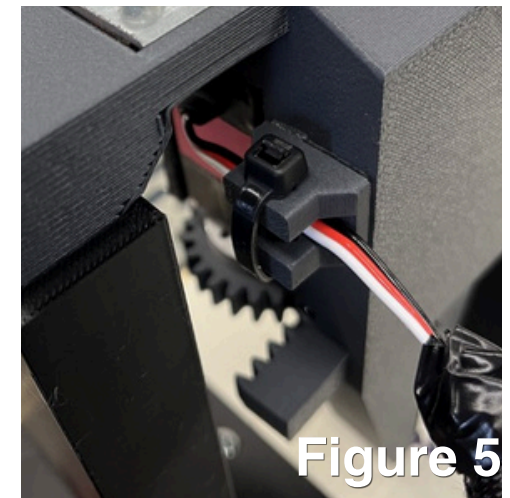
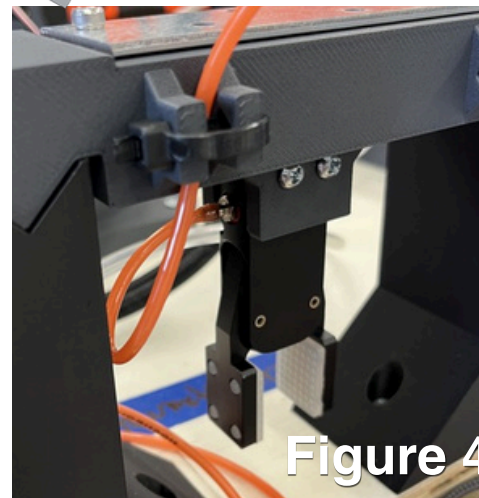
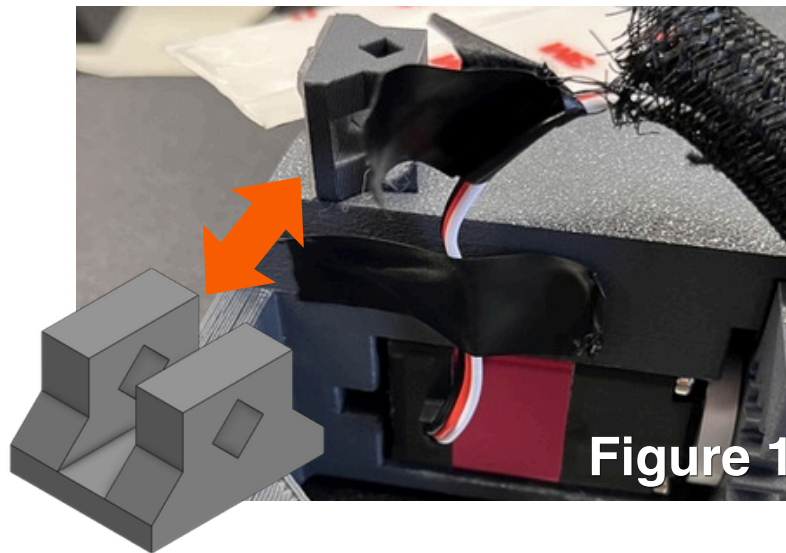
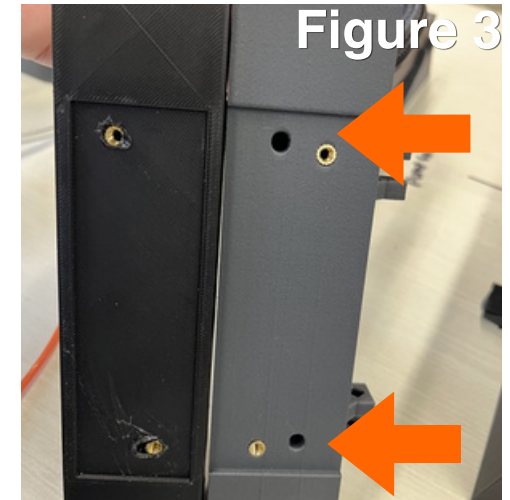
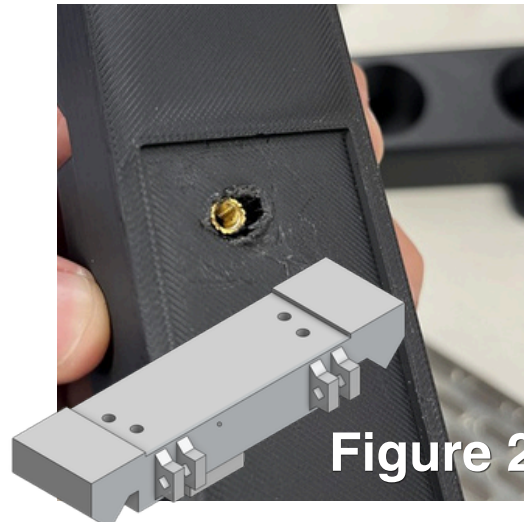
In Collaboration With:



Failure Analysis and Manufacturability Redesign

Root Cause Analysis

- Troubleshooting
- DFM
- Prototype iteration
- Reliability engineering
- Standardization
- Continuous improvement



In Collaboration With:



Cable Management

Industrial Design

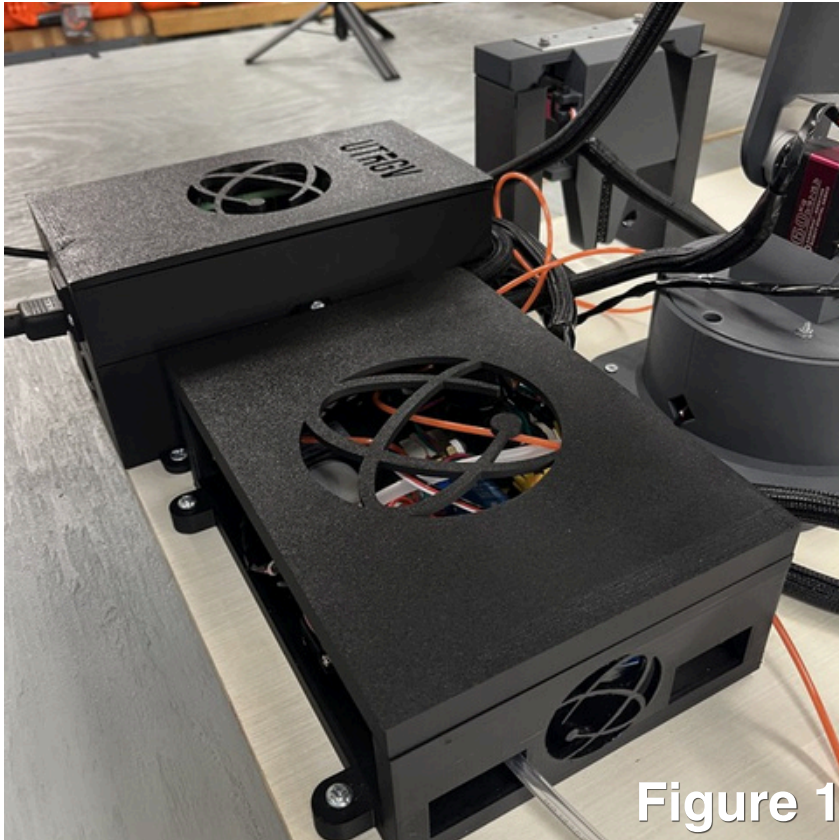


Figure 1

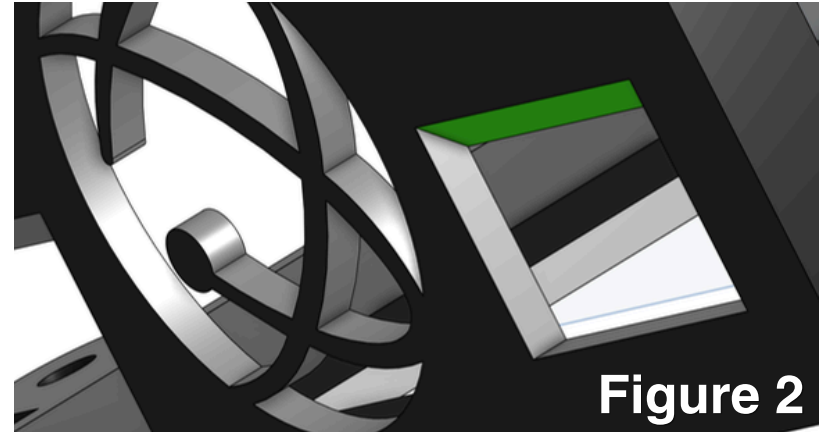
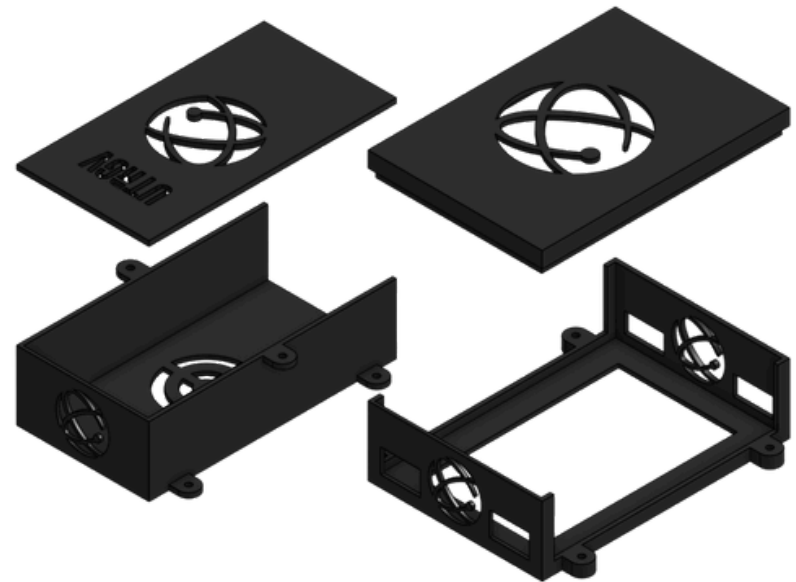


Figure 2



In Collaboration With:



Pickup Sequence

```
✓ GRIPPER_PICKUP_SEQ = [  
    {"move": [90, 90, 90, 90, 90], "pause_ms": 600},      # start home  
    {"move": [152, 90, 90, 90, 90], "pause_ms": 600},    # rotate base  
    {"move": [152, 56, 120, 120, 85], "pause_ms": 2000},  # straighten  
    {"move": [152, 102, 144, 152, 80], "pause_ms": 2000}, # get to latch position  
    {"relay": "MAGNET", "state": "ON"},                  # turn on magnet  
    {"move": [152, 102, 144, 152, 80], "pause_ms": 2000}, # confirm latch position  
    {"move": [152, 56, 144, 152, 80], "pause_ms": 2000}, # pull away / straighten  
    {"move": [90, 90, 90, 90, 90], "pause_ms": 600},    # home  
]
```

```
✓ ACTION_A_TASK_SEQ = [  
    {"move": [90, 90, 90, 90, 90], "pause_ms": 1000}, #HOME  
    {"move": [115, 90, 90, 90, 90], "pause_ms": 1000}, #HOME  
    {"move": [109, 90, 90, 90, 90], "pause_ms": 1000}, #HOME  
    {"test_servo": 40, "pause_ms": 5000}, #OPEN GRIPPER  
    {"move": [109, 110, 140, 62, 90], "pause_ms": 6000}, #HOVER  
    {"move": [109, 116, 150, 62, 90], "pause_ms": 5000}, #LOWER DOWN TO TARGET  
    {"test_servo": 67, "pause_ms": 5000}, #CLOSE GRIPPER  
    {"move": [109, 105, 120, 62, 90], "pause_ms": 4000}, #PICK UP FOR CLEARANCE  
    {"move": [60, 105, 120, 62, 90], "pause_ms": 6000}, #BASE CENTER  
    {"move": [75, 105, 120, 62, 90], "pause_ms": 6000}, #BASE CENTER  
    {"move": [75, 105, 145, 62, 90], "pause_ms": 4000}, #HOVER LOWER DOWN  
    {"move": [75, 110, 160, 62, 90], "pause_ms": 4000}, #LOWER DOWN  
    {"test_servo": 40, "pause_ms": 6000}, #OPEN GRIPPER  
    {"move": [75, 114, 120, 52, 90], "pause_ms": 1000}, #PULL UP  
    {"move": [90, 90, 90, 90, 90], "pause_ms": 1000}, #HOME  
    {"test_servo": 90, "pause_ms": 5000}, #CLOSE GRIPPER  
]
```

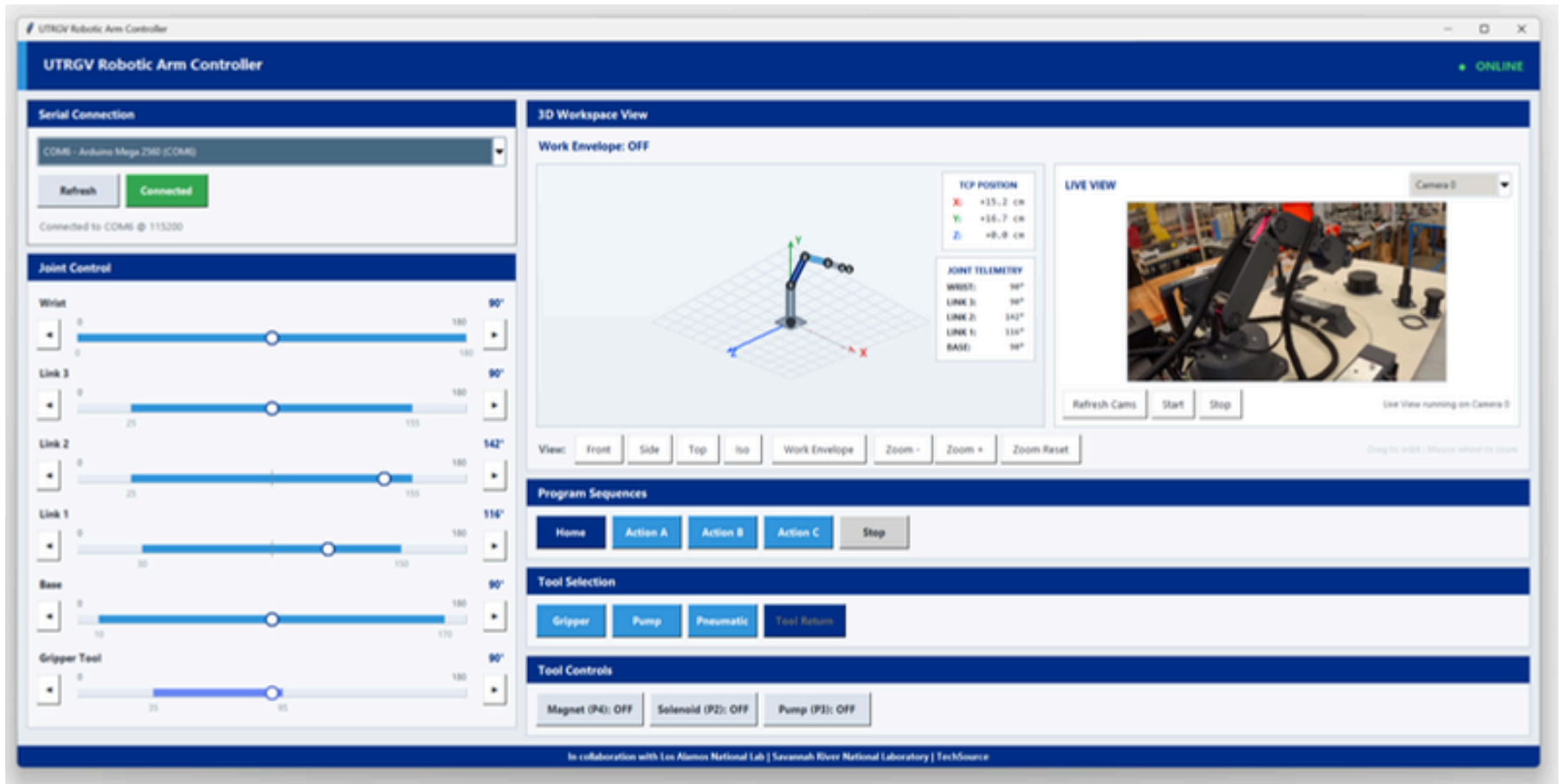
Task Sequence

```
✓ WORK_MIN = {  
    "base": 10,  
    "link1": 30,  
    "link2": 25,  
    "link3": 25,  
    "tool": 0,  
    "test": 35,  
}  
✓ WORK_MAX = {  
    "base": 170,  
    "link1": 150,  
    "link2": 155,  
    "link3": 155,  
    "tool": 180,  
    "test": 95,  
}
```

Servo Limits

GUI interface

UI Design



In Collaboration With:



GUI interface

State-Based Logic

Serial Connection

COM6 - Arduino Mega 2560 (COM6)

Refresh

Connected

Connected to COM6 @ 115200

Joint Control

Wrist 90°
0 180
0 180

Link 3 90°
0 180
25 155

Link 2 90°
0 180
25 155

Link 1 90°
0 180
30 150

Base 90°
0 180
10 170

Gripper Tool 90°
0 180
35 95

Program Sequences

Home

Action A

Action B

Action C

Stop

Tool Selection

Gripper

Pump

Pneumatic

Tool Return

Tool Controls

Magnet (P4): OFF

Solenoid (P2): OFF

Pump (P3): OFF

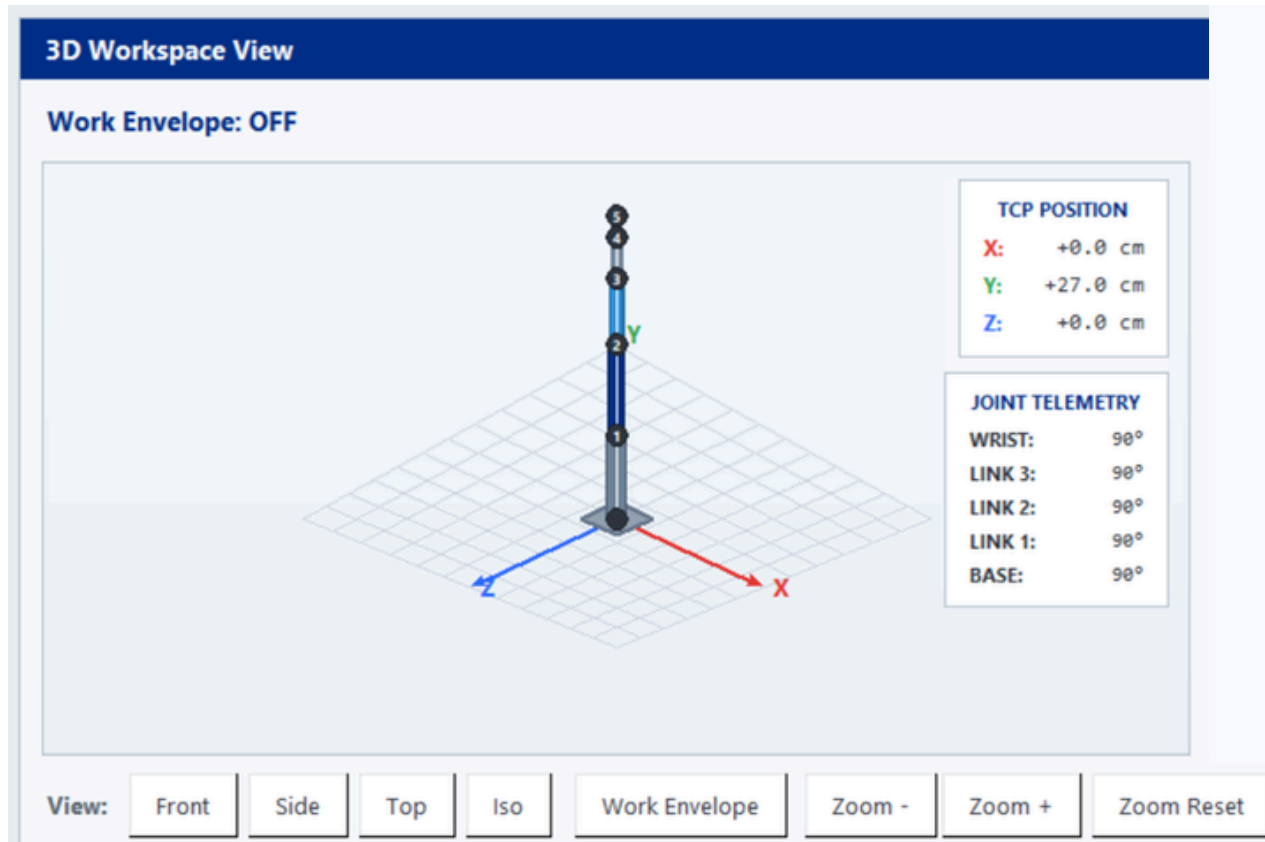
Program Sequences

In Collaboration With:



GUI interface

Data Visualization



JOINT TELEMTRY

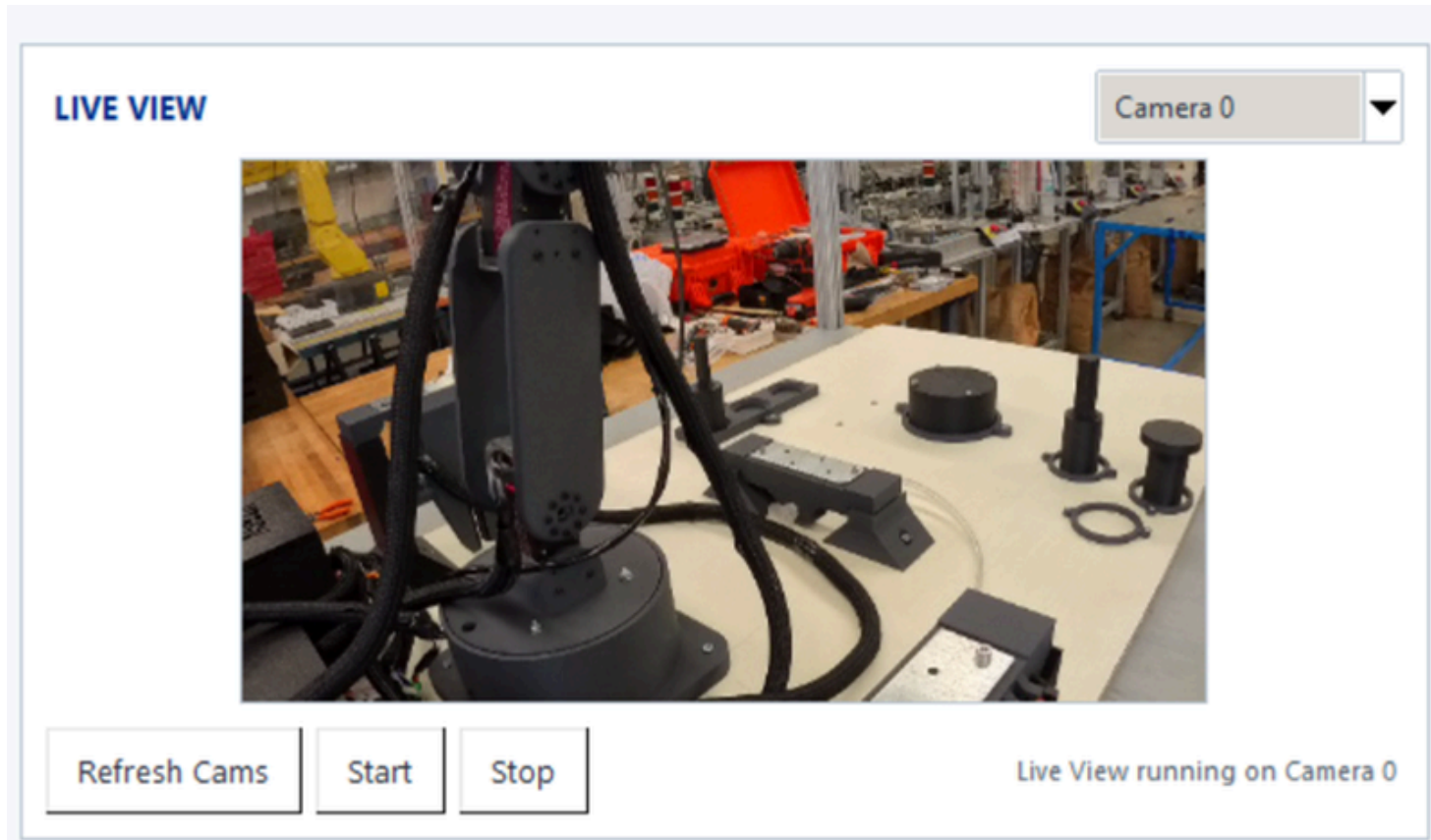
WRIST:	90°
LINK 3:	132°
LINK 2:	122°
LINK 1:	125°
BASE:	90°

In Collaboration With:



GUI interface

Camera Integration



In Collaboration With:



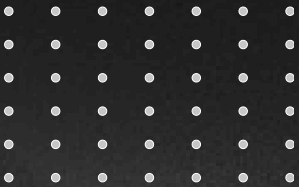


RoboLink X5

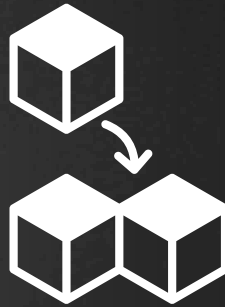
Great for research and
education

\$1500

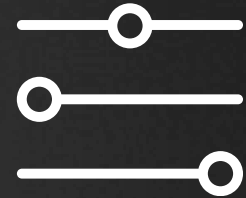
5 Degrees of Freedom



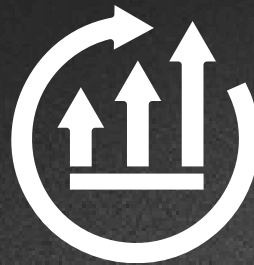
Core Features



Modular



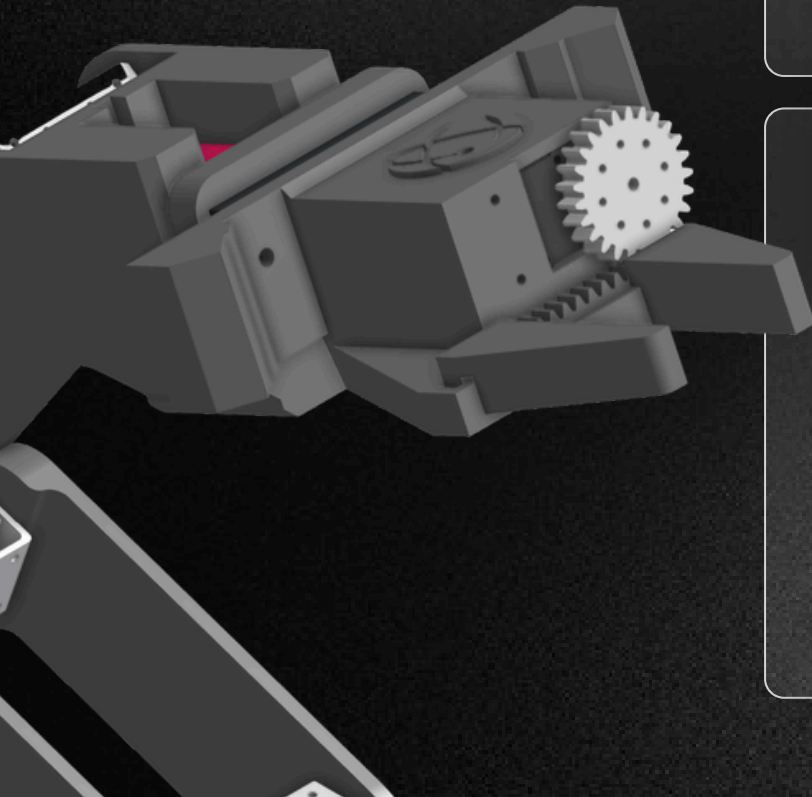
Customizable



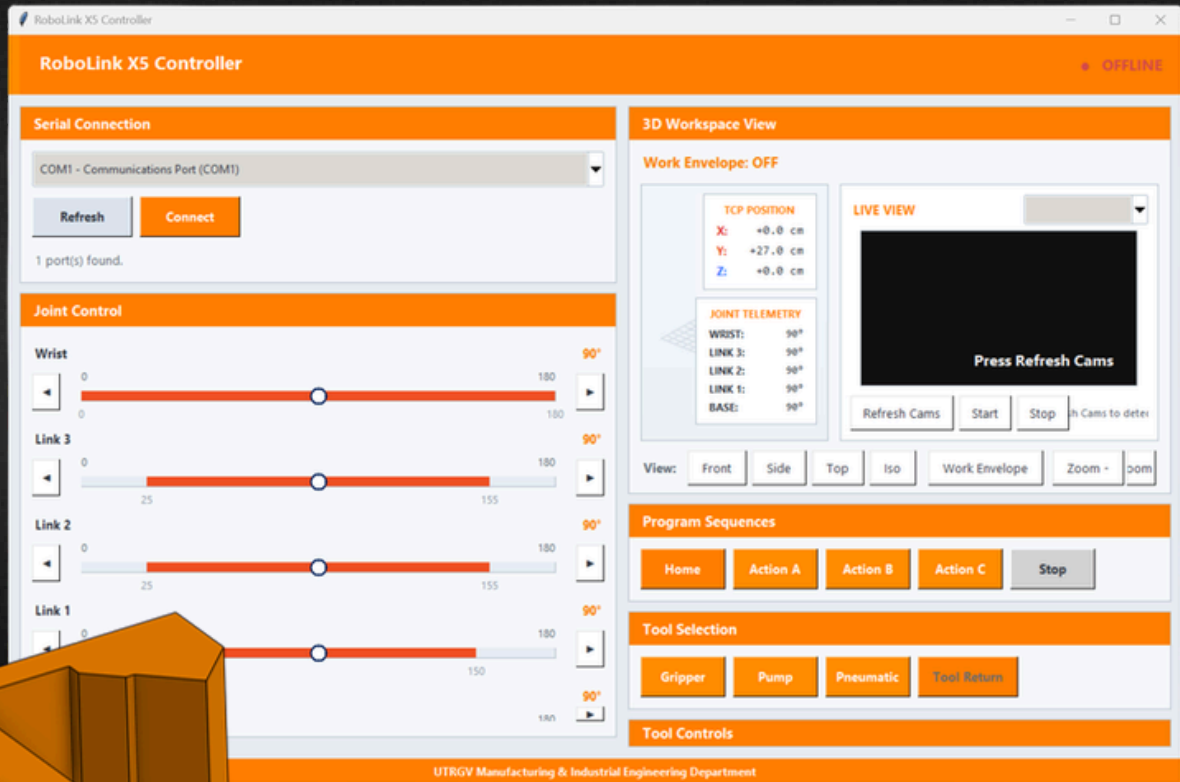
Scalable



Accessible



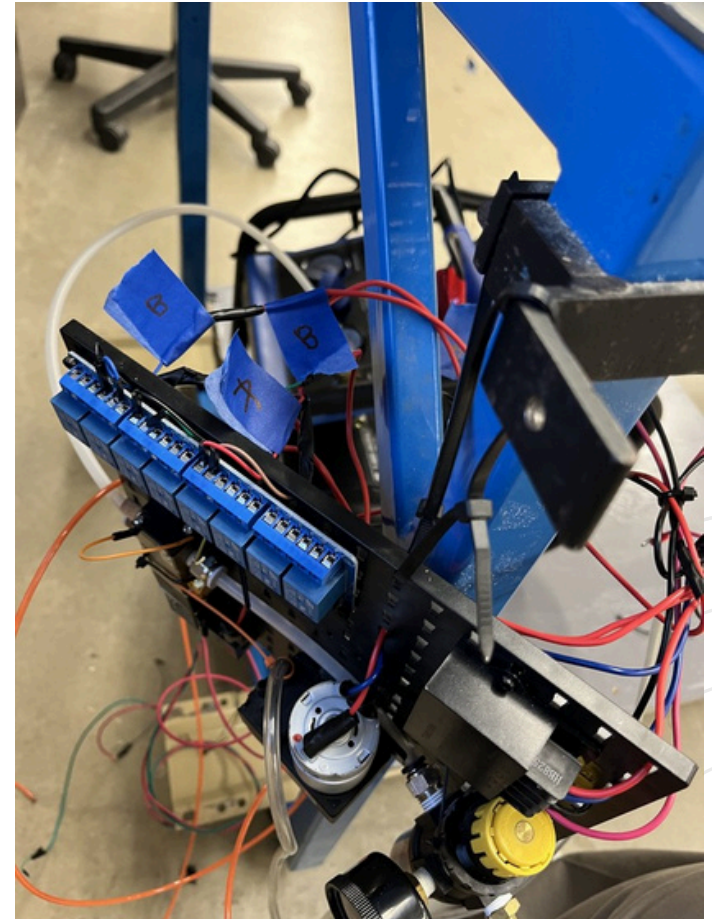
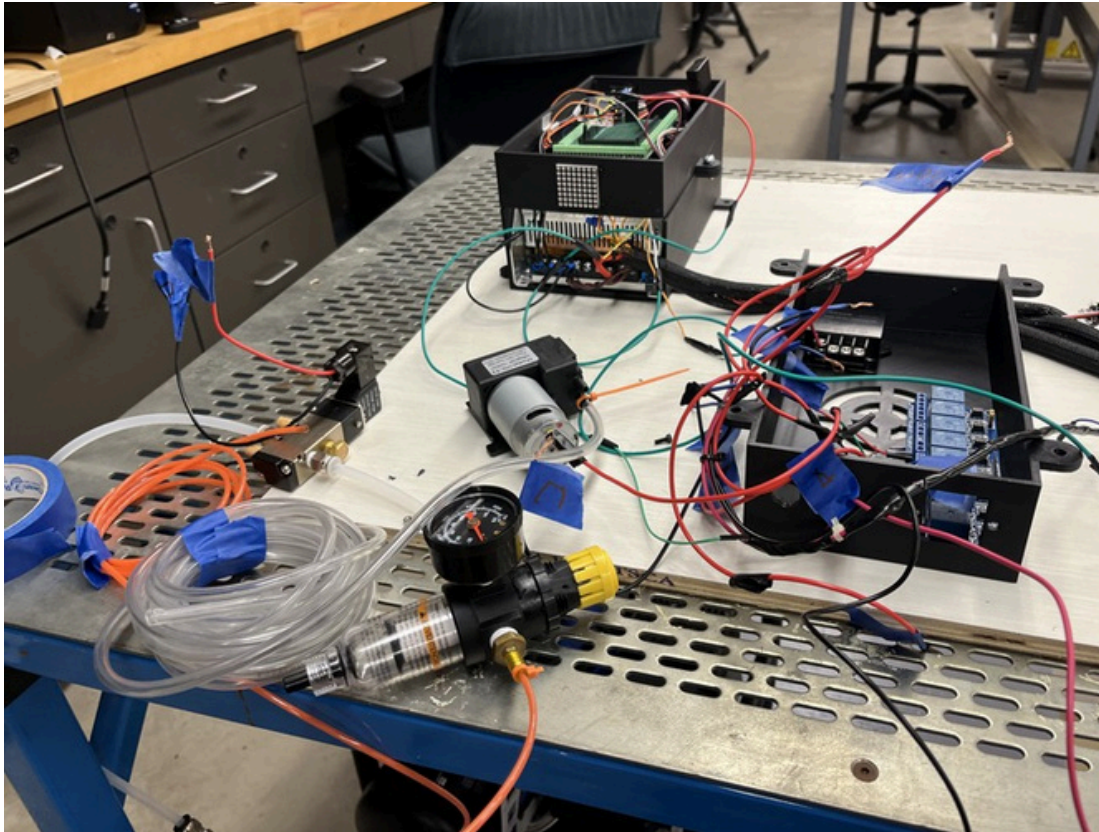
Your Brand



#000F7E	#172492	#212E9D	#2B3BA9	#3140B3	#515CBF	#7079CA	#999ED9	#C2C4E8	#E7E8F6
#002F71	#004098	#0053A5	#0070C1	#3296DC	#69C3FF	#96D5FF	#C3E7FF	#E1F3FF	#F5F8FF
#08412A	#126C48	#1B885C	#26A070	#29B77D	#2CC486	#7DDEB7	#9EE8CA	#BDEFDB	#DDF3EA
#171A22	#373A43	#555962	#686C76	#91959F	#B0B5BF	#D4D8E3	#E4E8F3	#EDF2FD	#F4F9FF
#292C30	#44494F	#5F666F	#727984	#8E959E	#A6ABB2	#C5C9CD	#DADDE0	#EAE8EE	#F5F5F7
#0C0D17	#2D2E37	#4A4B55	#5D5E68	#848590	#A5A6B1	#CACBD7	#DEDFEA	#EBECF7	#F5F6FF

Lessons Learned

Problem Solving

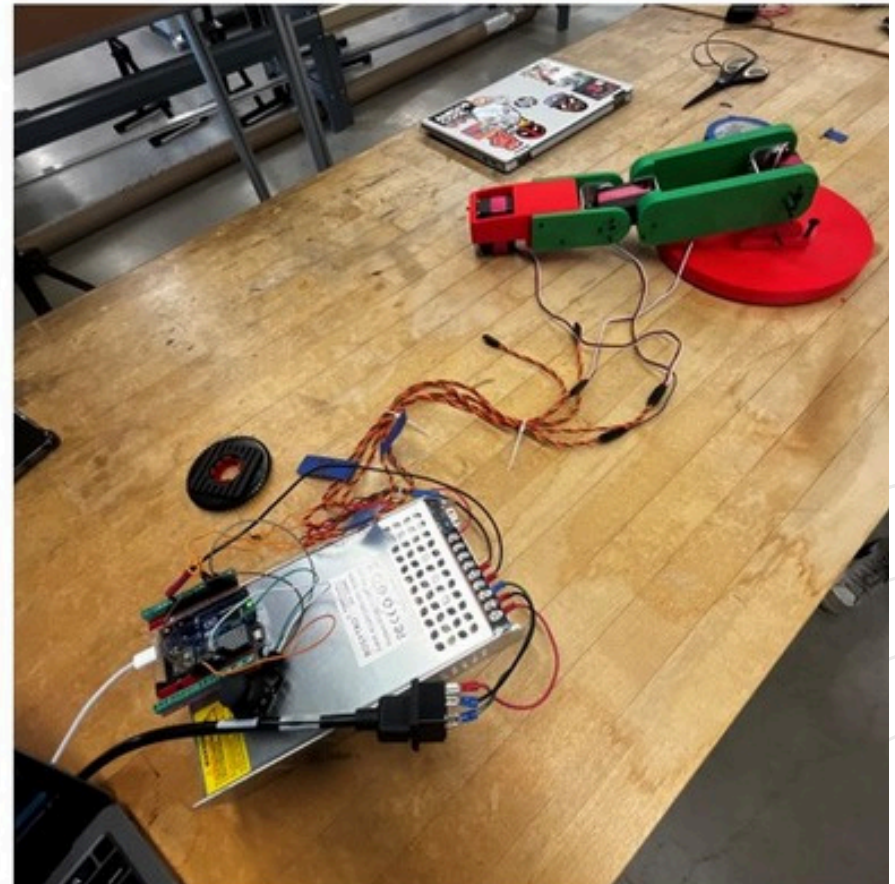


In Collaboration With:



Lessons Learned

Persistence



In Collaboration With:

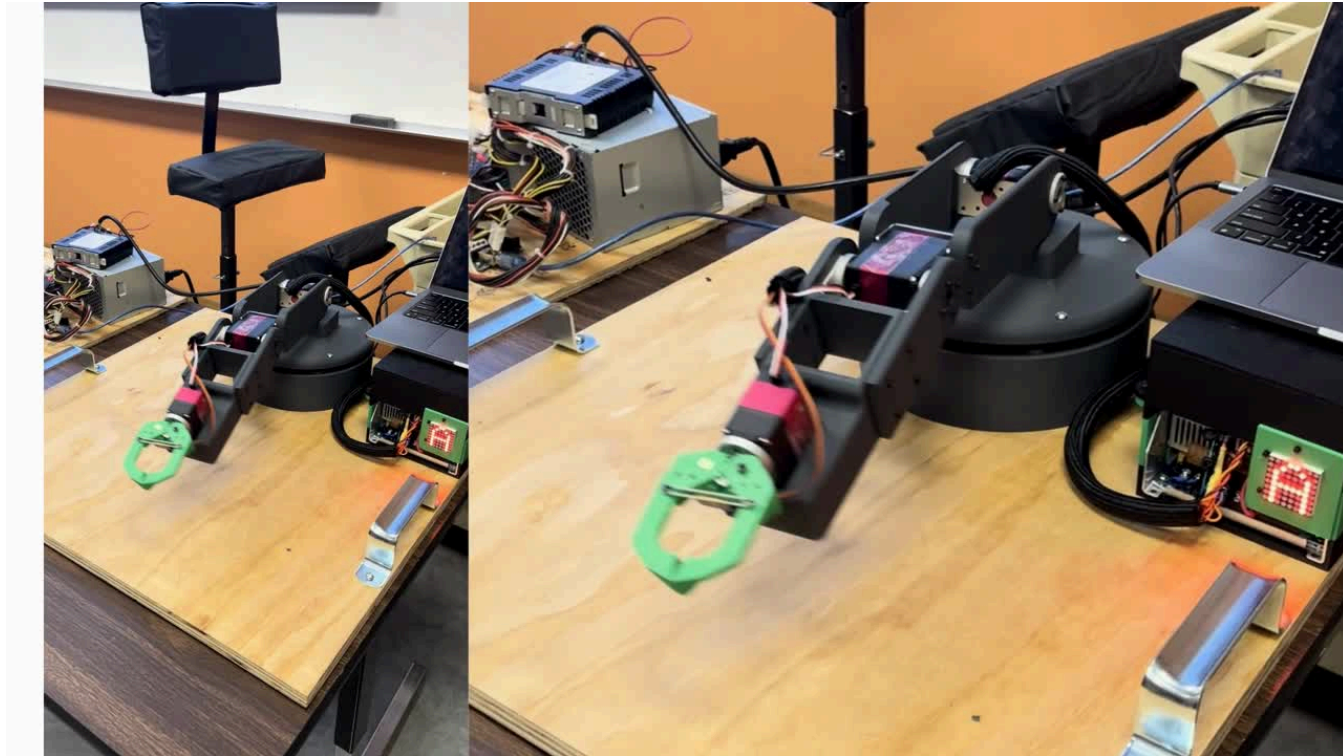


Prototype Timeline

Iterative Design



Early Prototype



December 2025

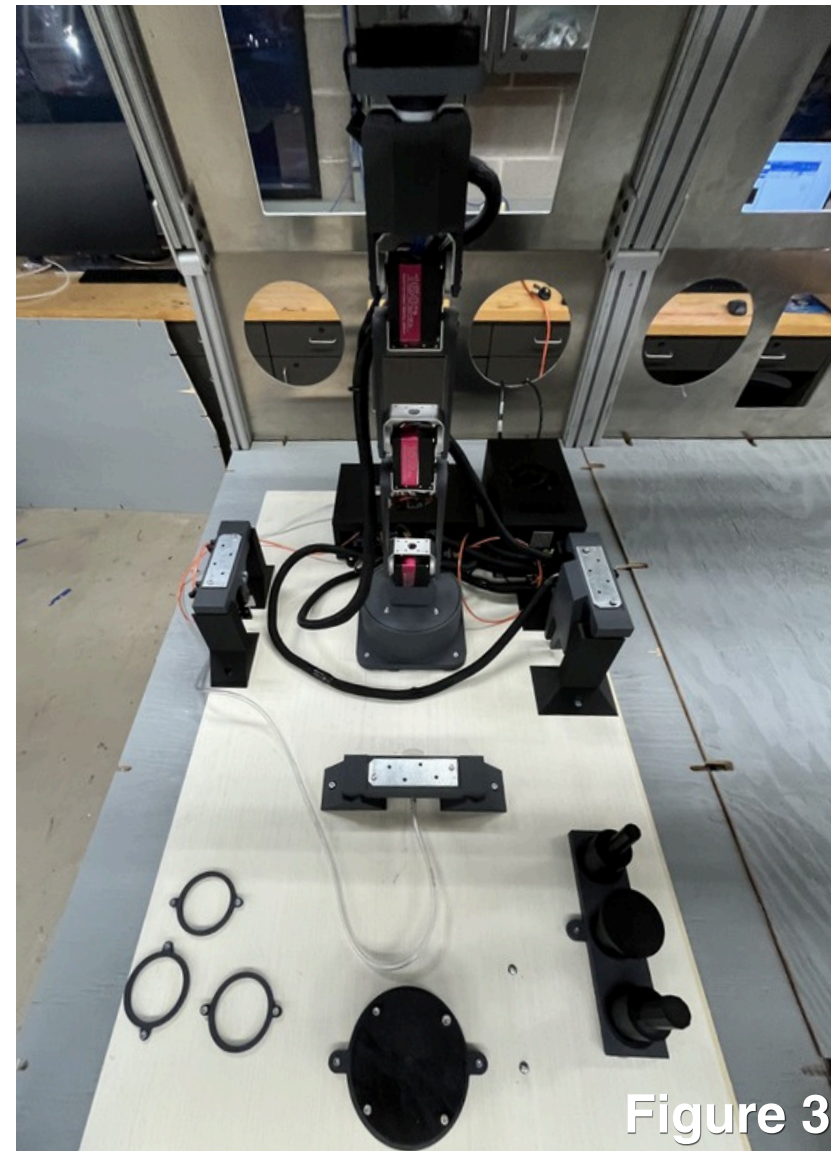
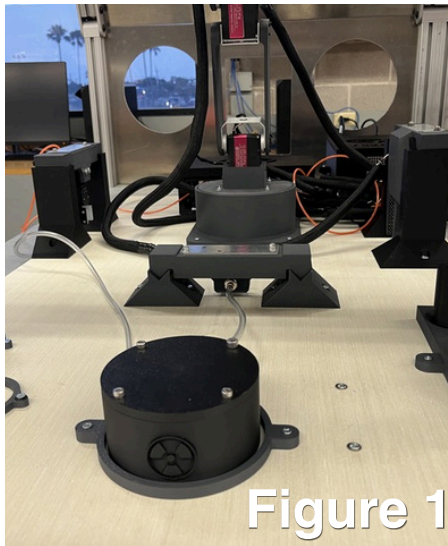
In Collaboration With:



Final Design Overview

Systems Engineering

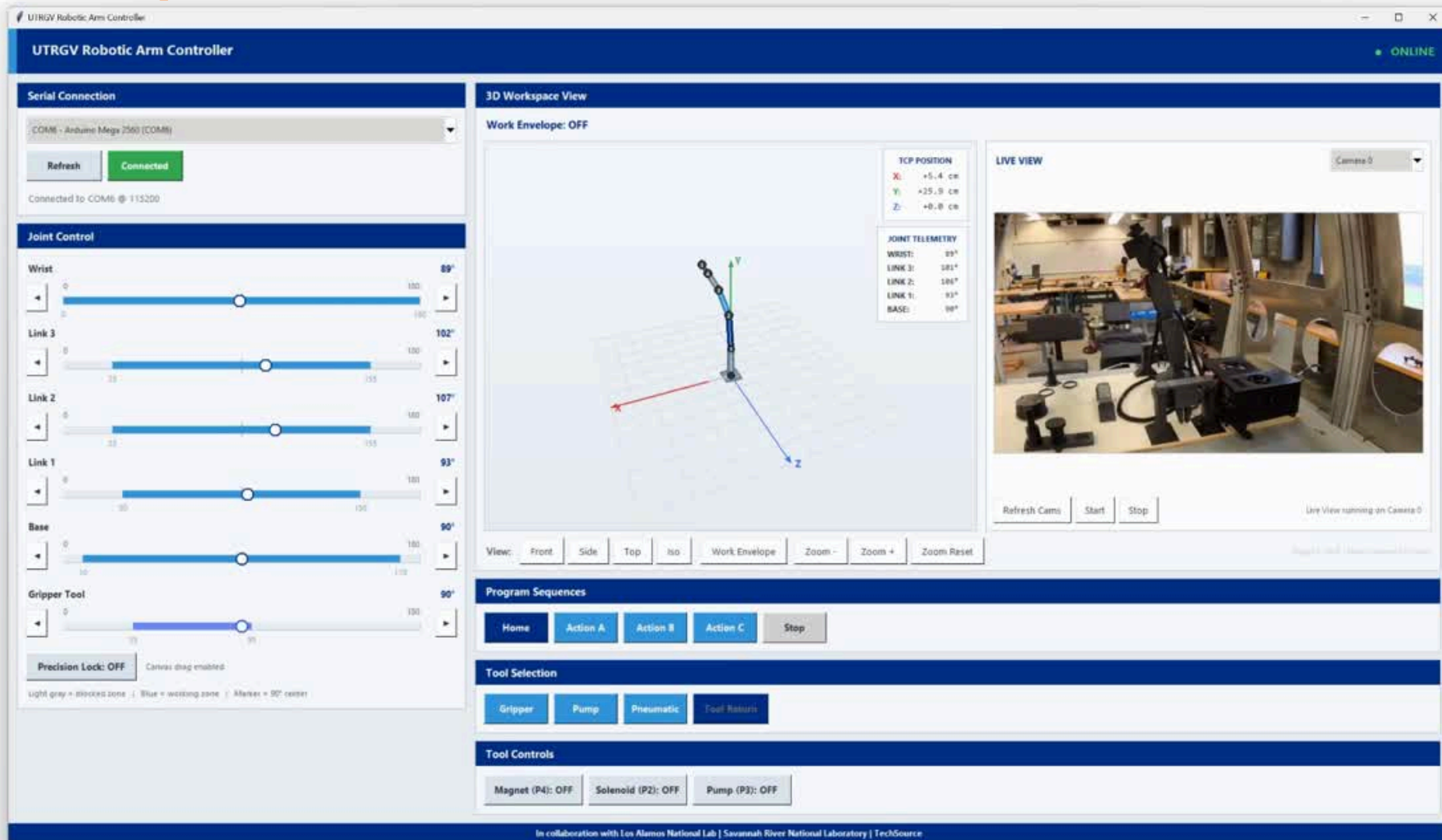
- 5 DOF Articulated Robotic Arm
- GUI
- Payload Test
- Tool Swapping Mechanism



In Collaboration With:



Graphical User Interface



In Collaboration With:



Teach Pendant

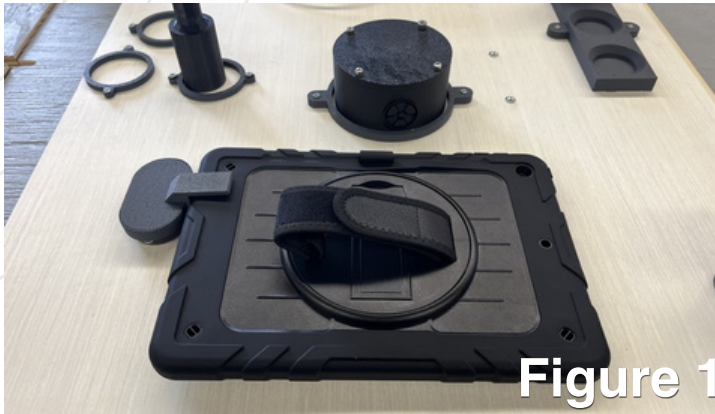
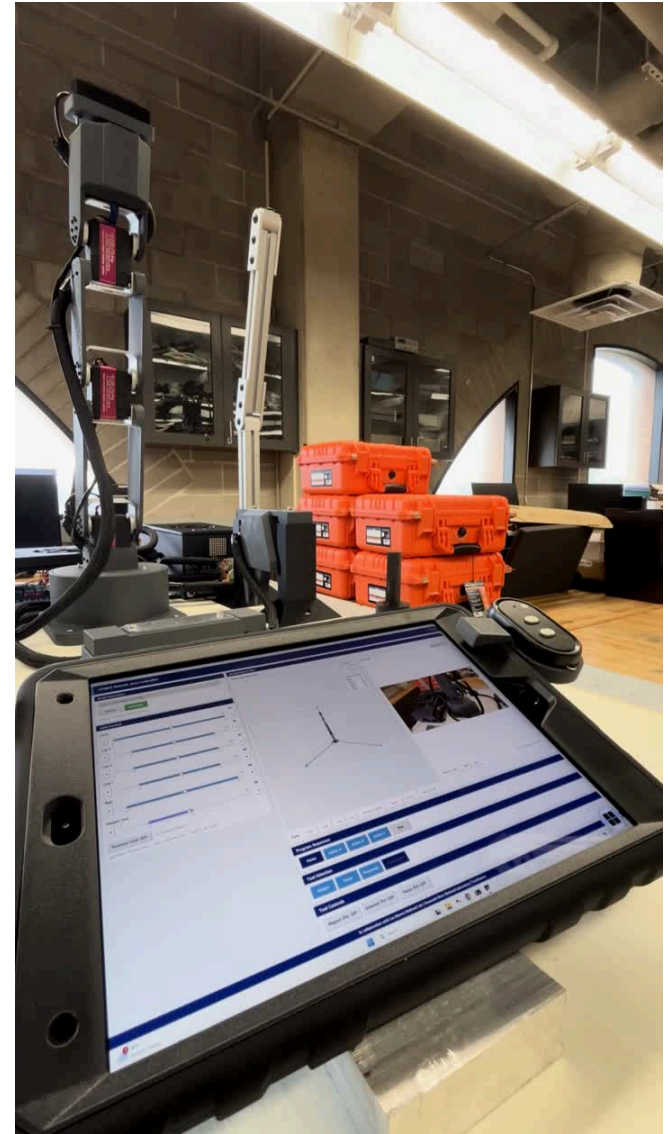


Figure 1



Figure 2



In Collaboration With:

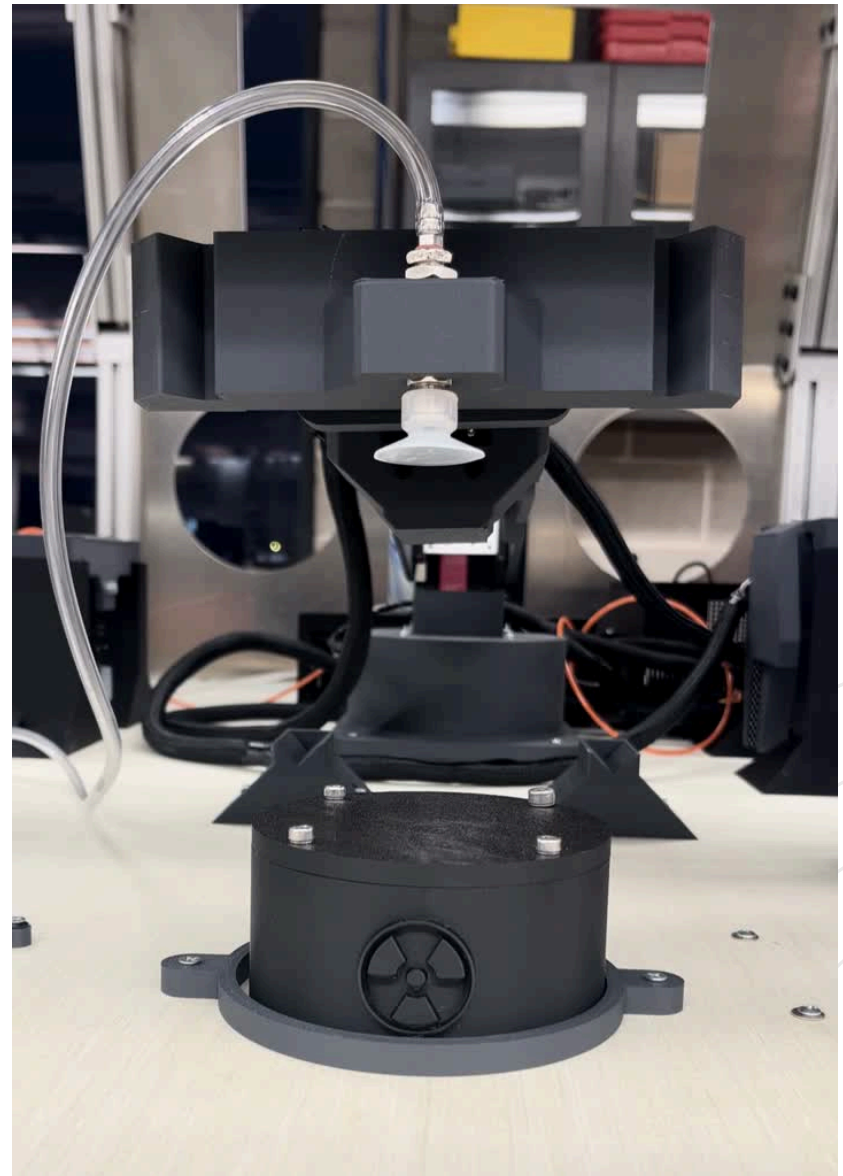


1KG Payload Test

Project Objective



Weight Verification



In Collaboration With:



Final Design Overview

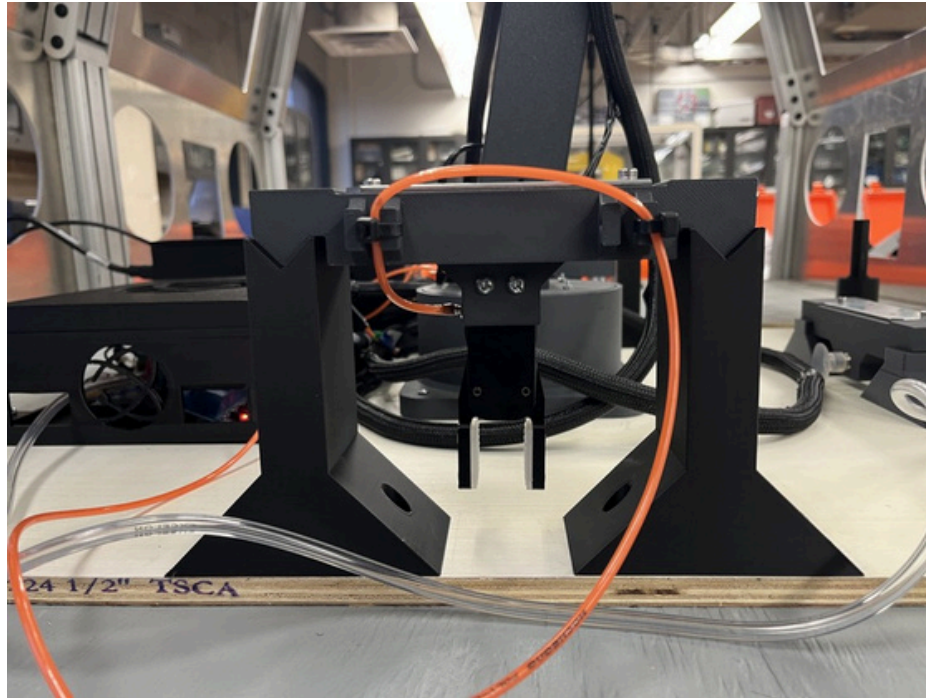


In Collaboration With:

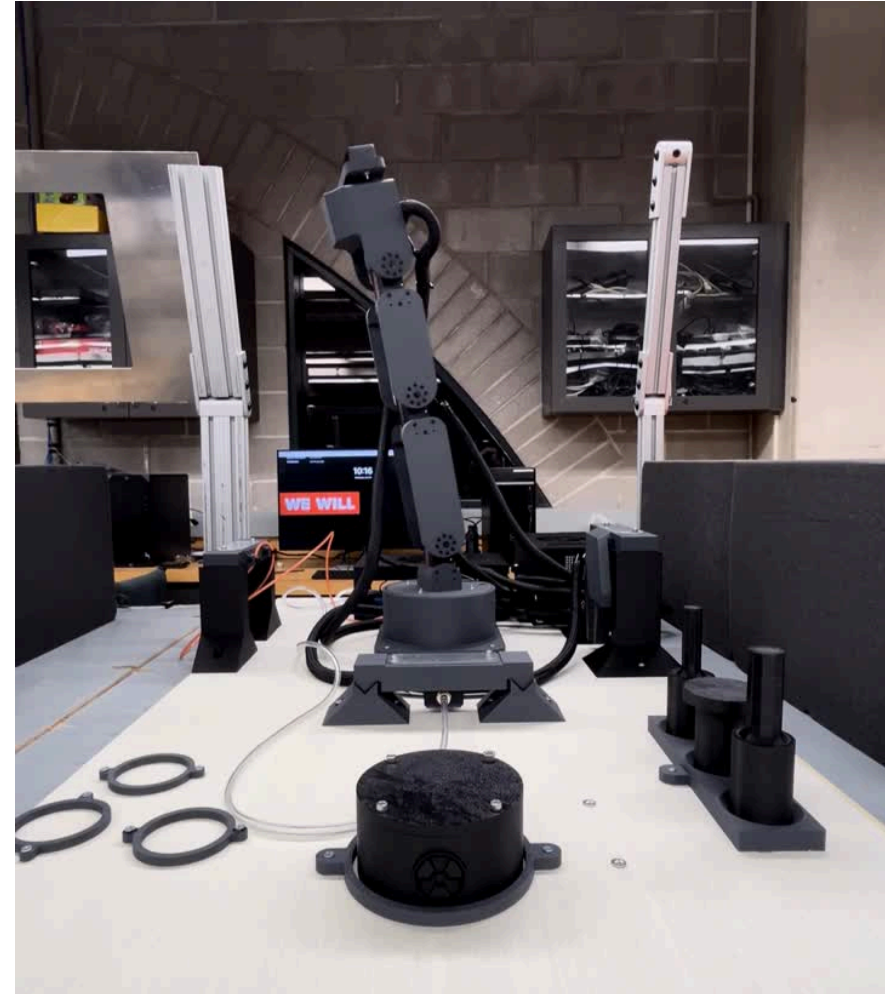


Automatic Tool Swapping Mechanism

Project Objective



Pneumatic Gripper

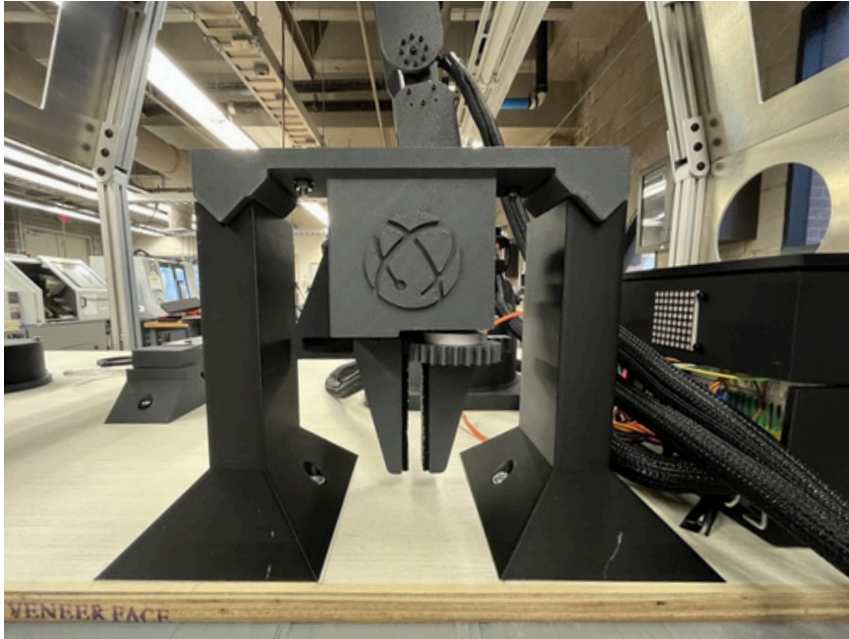


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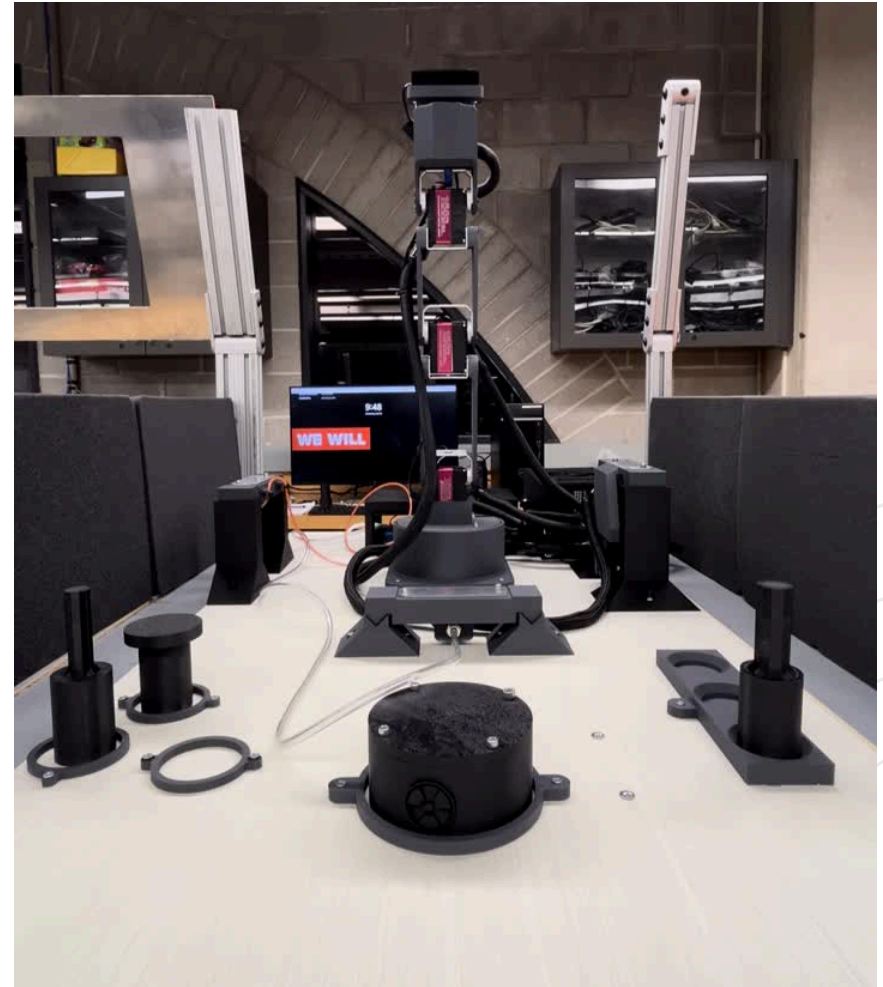


Automatic Tool Swapping Mechanism

Project Objective



Servo Gripper



In Collaboration With:

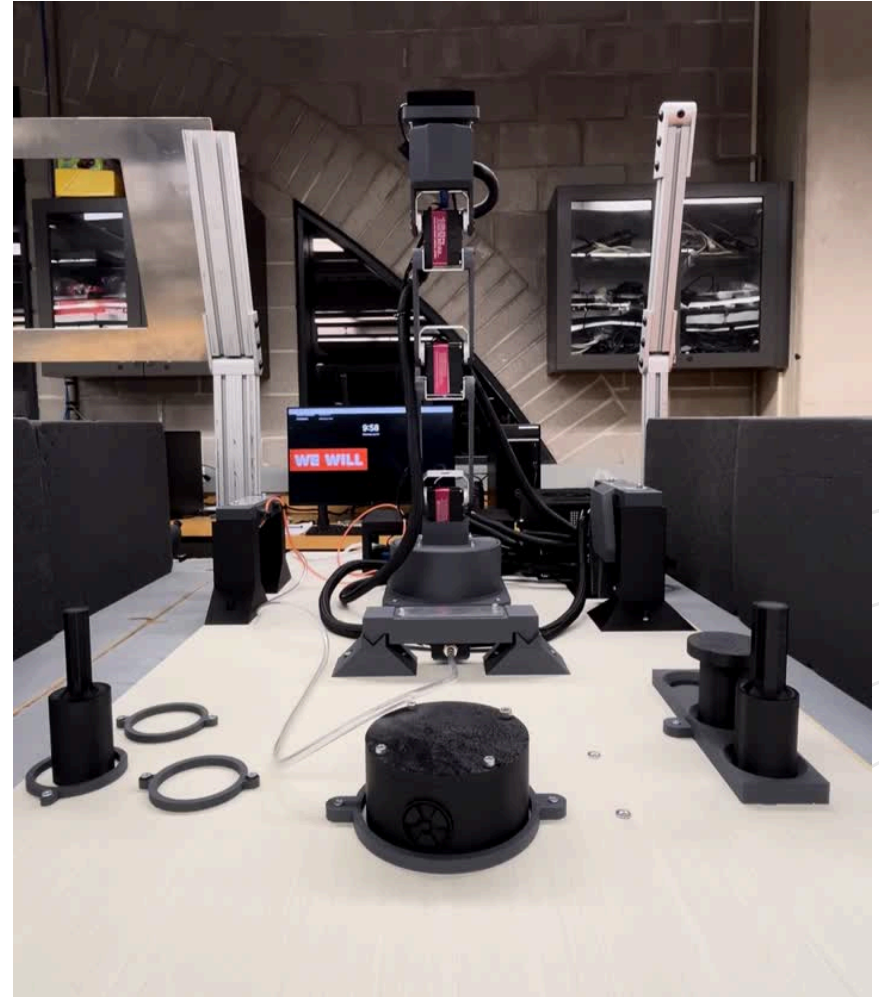


Automatic Tool Swapping Mechanism

Project Objective



Electric Pump



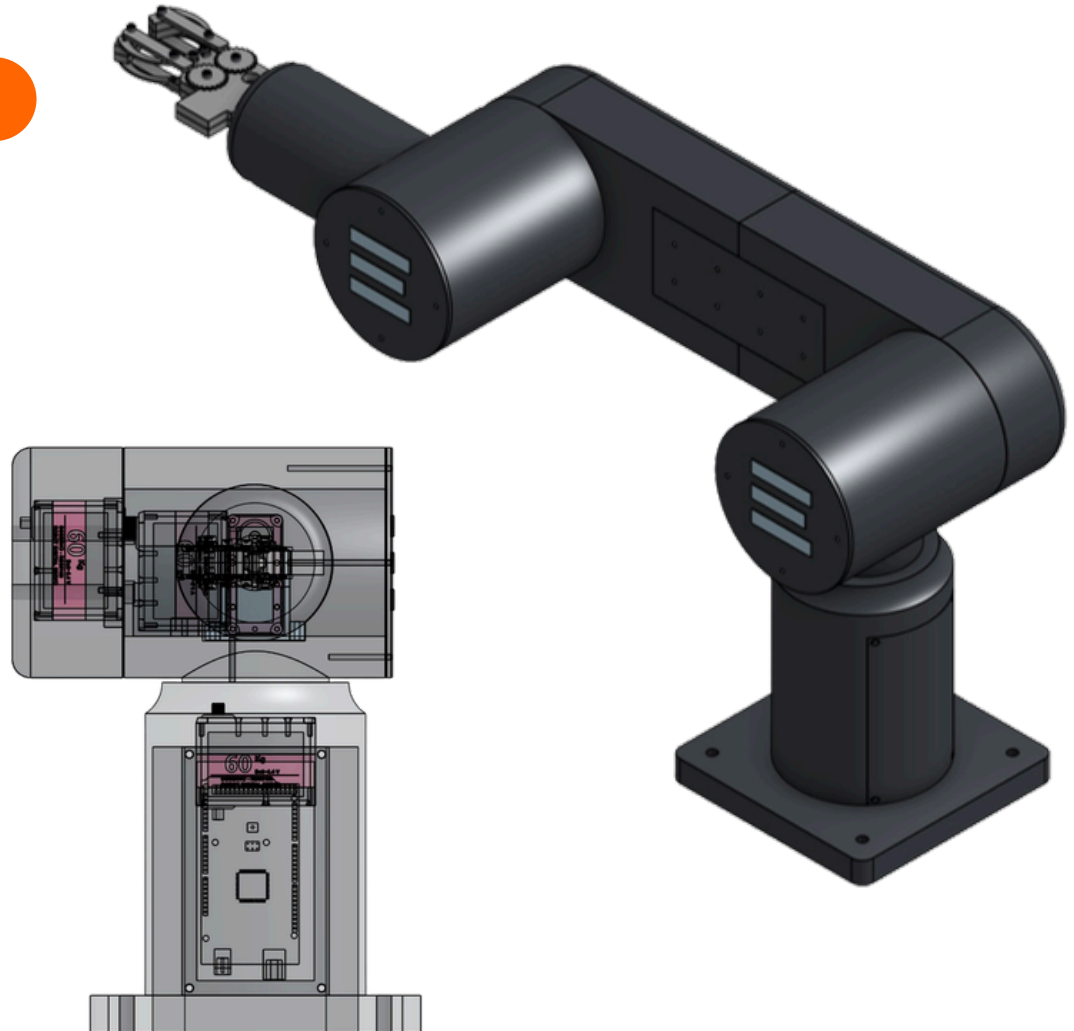
In Collaboration With:



Future Work

Senior Design

- IP Rating
- Machined Links
- Vision System
- Radiation-resistance
- AI Target Location
- Custom PCB
- Feedback System



In Collaboration With:



THANK YOU

Q&A

In Collaboration With:

